SAINT MICHAEL COLLEGE OF CARAGA

THE INTERPLAY OF 21ST-CENTURY COMPETENCIES AND LINGUISTIC COMPETENCE: A CORRELATIONAL ANALYSIS

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The Faculty of
College of Teacher Education

In Partial Requirements for the Degree
BACHELOR OF SECONDARY EDUCATION MAJOR IN ENGLISH

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SAINT MICHAEL COLLEGE OF CARAGA
College Of Teacher Education

CERTIFICATE OF RESEARCH APPROVAL

This thesis entitled **"**The Interplay of 21st-Century Competencies and Linguistic Competence: A Correlational Analysis**"** presented and submitted by BETHANY MAE P. MONTILLA in partial fulfillment of the requirements for the degree of Bachelor of Secondary Education Major in English is hereby accepted and recommended for Oral Examination.

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ABSTRACT

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I. OBJECTIVES

II. METHODOLOGY

III. FINDINGS

IV. RECOMMENDATIONS

KEYWORDS:

CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Education today requires more than knowing facts and completing tasks. Students must be able to communicate clearly, think carefully, work well with others, and create new ideas. These abilities, known as the 4Cs of 21st-century skills—communication, collaboration, critical thinking, and creativity—are now important for success in school, at work, and in daily life (Thornhill-Miller et al., 2023; Martínez & Bonilla, 2024)

At the center of these skills is linguistic competence, or the ability to use language correctly and effectively. Linguistic competence includes phonology (the sounds of language), vocabulary, word formation, sentence structure, meaning, and correct spelling. Students who have weak control of these areas often find it difficult to understand lessons, express ideas, solve problems, or participate in group activities (Qizi, 2024; Mytsenko, 2024). Strong language skills, on the other hand, help students speak clearly, organize their thoughts, and interact confidently with others.

Many international studies show that language skills and 21st-century skills are closely connected. Ünaldı & Yüce (2021) explained that understanding grammar and forming correct sentences helps students think more critically. Susoy (2024) found that students who analyze information well also perform better in reading. Other researchers observed that practicing language improves accuracy and adaptability Alkhrisheh et al. (2021), while the creative use of words supports innovation and problem-solving Sastromiharjo & Hadianto (2024). Shakirova et al. (2019) added that combining language practice with critical thinking increases creativity. Studies by Shadiev & Wang (2022) and Mytsenko (2024) also showed that strong language

proficiency allows learners to communicate and collaborate across cultures. Together, these findings suggest that developing linguistic competence naturally strengthens the 4Cs.

Philippine studies support the same idea. (Cabico & Sales, 2023) reported that grammar proficiency improves fluency and teamwork. Galang et al. (2024) found that English proficiency predicts better performance in Purposive Communication. Caldoza et al. (2024) showed that reading comprehension helps students think more critically and reflect on what they learn. Azarcon-Sanchez et al. (2024) noted that English teachers actively promote creativity and critical thinking in their classes. Herradura Basilan & Paula Flor (2024) found that communicative activities help build collaboration among students, while Kadatuan et al. (2024) confirmed that communicative competence improves when learners participate actively in language tasks.

Even with these findings, many colleges still have difficulty connecting language learning with the development of the 4Cs. Several studies noted that language subjects are often taught separately from activities that build communication, collaboration, creativity, and critical thinking (Pardede, 2020; Caldoza et al., 2024). Because of this, students may complete written requirements but still struggle to explain their thoughts, work with peers, or produce original outputs. This situation shows a gap between what students learn in language classes and how they apply these skills in real-life learning tasks (Azarcon-Sanchez et al., 2024; Mae et al., 2024)

A similar concern can be seen at Saint Michael College of Caraga (SMCC), especially among first-year students from the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS). Some students have basic English knowledge but find it hard to express ideas clearly, organize their thoughts, or participate actively in group work. Others can work alone but struggle when asked to think deeply, communicate complex ideas, or create original outputs. These challenges suggest that their level of linguistic competence may be affecting how well they develop the 4Cs.

Although many international and local studies have explored language skills and 21st-century skills**,** very few have examined how specific components of linguistic competence—such as phonology, vocabulary, grammar, and meaning—relate to the 4Cs among first-year Filipino college students. No study has yet focused on this relationship within the context of SMCC, where concerns about communication, collaboration, creativity, and critical thinking continue to appear in classroom situations.

Because of this gap, it is important to investigate how linguistic competence influences the 4Cs among SMCC first-year college students.

This study aims to examine the relationship between students' linguistic competence and their communication, collaboration, creativity, and critical thinking skills. The findings may guide teachers and school administrators in improving teaching strategies, learning activities, and curriculum design. By strengthening both language development and 21st-century skills, the study hopes to help prepare students to become effective communicators, critical thinkers, team players, and creative problem-solvers. Ultimately, this research contributes to the goal of producing globally competent and linguistically skilled Filipino graduates.

1.2 Review Related Literature

This chapter examines new studies on language skills— phonology, syntax, vocabulary, semantics, morphology, and spelling—as well as the 4Cs of 21st-century skills: critical thinking, communication, collaboration, and creativity. It focuses on research that shows how language knowledge connects with the skills needed in school and work today. Many studies also show how technology helps both language learning and the 4Cs.

Linguistic Competence

Linguistic competence means knowing a language well and using it correctly. It is an important part of communicative competence, which includes grammar, vocabulary, pronunciation, and how people use language in real life (Abdel Wahed Shaalan & Ahmad, 2024; Wahyuni et al., 2022). This skill helps learners understand and express meaning clearly in school, social, and work situations.

Research shows that linguistic competence includes grammar, pronunciation, and vocabulary (Świątek et al., 2023). These help learners make correct sentences and communicate properly. A study on Thai students learning English found that pronunciation, meaning, and sentence structure are important for listening and understanding (Singhkum & Patanasorn, 2022). It means teachers should include these skills in their lessons.

In education, linguistic competence helps students improve their communication skills. Y. Wang & Xie (2022) said that both linguistic and discourse competence help students write better. Younis & Shah (2023) also said that textbooks made for communicative learning help students improve their language skills. Mao (2023) added that students should learn how to use language in real-life situations.

Some studies also show that a person's background and environment affect their language skills. Davydova (2022) found that social identity and language background influence how students learn English. Sebahat Gözüm et al. (2020) also found that effective communication is important across all jobs, even in healthcare, where clear communication affects service quality. It means language learning is not only about grammar but also about culture and context.

However, there are still a few studies about how linguistic competence develops in different learners. Chan et al. (2023) found that bilingual children's language skills develop together, suggesting the need for long-term studies. Yu & Dong (2022) said that technology can help students improve their language skills through interactive learning. In the Philippines,

however, few studies connect linguistic competence with the 4Cs—communication, collaboration, critical thinking, and creativity—making this a good topic for future research.

In short, linguistic competence is the heart of language learning and communication. It helps connect theory and real-life use of language. To make learning more effective, linguistic competence should be studied alongside communication, technology, and culture to meet today's language needs.

Phonology

Phonology is a core component of linguistic competence, encompassing both segmental features and suprasegmental features such as stress, rhythm, and intonation. Explicit phonological instruction significantly enhances learners' ability to perceive and produce sounds accurately. For example, Shukurullayeva & Qizi (2024) found that phonetics-based instruction meaningfully improves pronunciation and sound discrimination, reinforcing the need for phonology-informed pedagogy. This is supported by Suzukida & Saito (2022) who demonstrated that segmental and suprasegmental features jointly shape L2 English pronunciation, indicating that phonological competence contributes not only to sound production but also to overall communicative effectiveness.

Building on this foundation, research on phonological awareness (PA) highlights its role in early literacy development. PA enables learners to identify and manipulate sound units such as syllables and phonemes, skills essential for decoding and reading. In the Philippine context, Abdon and Abdon & Barrios (2022) showed that a structured mother-tongue PA intervention significantly improved kindergarten learners' phonological awareness, letter knowledge, and even early English reading abilities. These findings support Cummins's Linguistic Interdependence Hypothesis, suggesting that strengthening phonological skills in L1 can positively transfer to L2

literacy. This underscores the importance of mother tongue–based multilingual education, especially in linguistically diverse communities.

As literacy-related aspects of phonology develop, research on pronunciation emphasizes the importance of integrating both segmental and suprasegmental instruction. While learners often improve segmental accuracy more quickly, suprasegmental features play a crucial role in intelligibility and fluency. Luchini & Kickhöfel Alves (2025) found that suprasegmental instruction enhances prosodic awareness and contributes to more holistic pronunciation gains. Similarly, in a study of adult EFL learners, Rodríguez et al. (2025) reported that improvements in rhythm, stress, and intonation increased learners' fluency, expressiveness, and overall communicative confidence. This aligns with Aguion (2022) call for an intelligibility-oriented approach to pronunciation teaching in the Philippines, emphasizing communicative clarity over native-like accuracy.

As pronunciation pedagogy evolves, technological advancements have expanded its possibilities. Automatic Speech Recognition (ASR) systems now offer real-time, objective feedback that supports sustained pronunciation practice. Sun (2023) found that ASR-based training improved L2 learners' speaking performance by encouraging self-correction and iterative learning. Complementing this, a meta-analysis by Ngo et al. (2024) reported that ASR tools produce moderate and consistent improvements in pronunciation, particularly when paired with explicit corrective feedback. Moving toward more advanced models, Sungkur & Shibdeen (2025) developed a generative AI–enhanced pronunciation trainer that provides instantaneous feedback through ASR and TTS integration, showing strong potential for personalized and multilingual instruction. Duris et al. (2025) similarly argue that AI technologies enable scalable and

autonomous pronunciation practice, an idea supported by Liang & Fryer (2024) empirical findings that learners effectively engage with ASR tools in independent learning contexts.

Alongside these technological advancements, scholars have increasingly recognized that phonology interacts with other linguistic subsystems in shaping literacy and language development. Kim (2023) demonstrated that phonological awareness, morphological awareness, and orthographic pattern recognition collectively account for substantial variance in reading performance. This is echoed in findings by Zhang et al. (2021) who reported that morphological awareness supports learners' ability to activate phonological and orthographic representations in L2 reading. Furthermore, Li & Chen (2024) observed that learners' reliance on orthographic or phonological cues varies by proficiency, with more advanced readers integrating both systems fluidly. Intervention studies by Shen et al. (2024) also show that training in phonological and morphological skills strengthens reading outcomes, while cross-script research such as Young-Scholten et al. (2024) illustrates that L1 orthographic systems can shape L2 phonological development. Together, these findings reinforce the importance of integrated approaches to literacy instruction.

These interactions are especially visible in multilingual environments such as the Philippines, where L1–L2 transfer shapes learners' phonological development Sales (2022) for example, found that Cebuano-speaking junior high students experience difficulty with English vowels that do not align with Cebuano phonology. Sales (2022) documented similar transfer effects among Hiligaynon speakers, including both segmental and suprasegmental influences on English pronunciation. Beyond error-based perspectives, Dumanig et al. (2025) observed that some non-target-like phonological patterns among Cebuano-Visayan learners can aid meaning-making in academic discourse, illustrating the complex role of phonology in multilingual

communication. Prosodic variation across Philippine English varieties also reflects strong L1 influence. Dumanig et al. (2025) found that acrolectal, mesolectal, and basilectal sociolects of Philippine English differ along a stress-timed to syllable-timed continuum. Teacher-related factors likewise shape phonology instruction, as Dumalig & Bayan (n.d.) reported that limited teacher training and low confidence in teaching suprasegmentals hinder effective classroom implementation.

Although much of the research on phonology centers on literacy and pronunciation, emerging scholarship suggests broader relevance to 21st-century skills, particularly communication, collaboration, critical thinking, and creativity. Prosodic mastery enhances intelligibility and supports clearer interpersonal communication, an essential element of the 4Cs (Rodríguez et al., 2025). ASR-based tools also provide opportunities for peer-supported learning, which strengthens collaboration (Ngo et al., 2024). Additionally, phonological awareness interventions—especially in L1—promote metalinguistic reflection, encouraging learners to analyze how sound systems operate across languages, thereby supporting critical thinking (Abdon & Barrios, 2022). Creative exploration is likewise fostered by AI-based pronunciation tools, such as those designed by Sungkur & Shibdeen (2025) which allow learners to experiment with pitch, rhythm, and expressive variation.

Despite these promising directions, two significant gaps remain in existing research. First, little empirical work has investigated how phonological competence—especially suprasegmental proficiency—contributes to the development of 21st-century competencies among Filipino tertiary learners. Although some studies link prosody to communicative competence, its role in collaboration, critical thinking, and creativity has not been systematically examined. Second, although ASR and AI technologies show strong potential for pronunciation learning, few tools have

been designed to fit the linguistic realities of Filipino learners, particularly those involving L1 phonological systems, Philippine English prosody, and local sociolinguistic variation. Research on integrating such tools into higher education curricula also remains scarce.

Given these gaps, the present study—examining the relationship between phonological competence and the 4Cs among Filipino tertiary students—offers a meaningful and timely contribution. By connecting phonology with higher-order competencies and localizing technological approaches to pronunciation learning, the study can advance both theoretical understanding and pedagogical practice in Philippine higher education.

Syntax

Syntax refers to the rules and patterns that determine how words are arranged to form phrases and sentences in a language. It is a core part of linguistic competence because it enables learners to construct grammatically correct and meaningful sentences. Zhou et al. (2020) explain that syntax helps speakers organize words into logical expressions, thereby making communication more effective. In learning a second language, syntactic knowledge enables students to understand complex sentences and produce accurate, fluent writing and speech.

Many studies have shown that strong syntactic skills are important for reading, writing, and understanding. Pérez-Mayos et al. (2021) claimed that being aware of syntactic connections within sentences can help readers understand what they are reading better because it becomes easier to process meaning and find textual cohesion. T. Liu et al. (2023) claimed that being aware of syntactic connections within sentences can help readers understand what they are reading better because it becomes easier to process meaning and find textual cohesion. Y. Li & Risteski (2021) indicated that knowledge of syntax can improve writing accuracy and sophistication, thereby helping students better organize their thoughts. (Stengel-Eskin et al., 2021). Glavaš &

Vuli´cvuli´c (2020) state that grammatical and communicative flexibility might be increased with the help of syntactic awareness exercises, such as the identification of the limits of the clause, paraphrasing the sentences, or understanding the real text samples. A teacher can also strengthen syntactic development by providing feedback on sentence variability, grammatical correctness, and coherence. (K.-H. Huang & Chang; E. Yang et al., 2022).

Technology has also improved how syntax is taught. Studies by Sachan et al. (2020) and Ahmad et al. (2021) show that using visual tools to illustrate sentence structures helps students build metalinguistic awareness. Such tools enable self-study and a profound understanding of how syntax works in context-specific ways. These pedagogical inventions indicate that teachers can teach syntax as a structural, functional, and creative component.

Syntax is not only a linguistic structure but also a social and educational tool. Kuncoro et al. (2020) argue that syntactic proficiency enhances communicative competence by helping learners express subtle ideas in academic and professional settings. Likewise, Hutchison (2020) and Chida et al. (2020) emphasize that effective use of syntax supports higher-order skills such as argumentation, reasoning, and creativity in writing. Classroom instruction in syntax, therefore, does more than teach grammar—it also develops collaboration and critical thinking, two core elements of 21st-century learning.

Despite the growing body of international research, few Philippine studies examine how syntactic competence relates to the 4Cs—communication, collaboration, critical thinking, and creativity—particularly among college EFL learners. Future research should explore how syntactic instruction can be integrated into communicative and outcomes-based curricula to improve grammar, cognition, and overall linguistic performance.

In conclusion, grammar is a part of language and communication skills that affects how well students can read, write, and speak. There is evidence that pedagogical projects that include

sentence-combining and syntactic awareness practices help students understand and write better. Teachers can use these techniques to help students learn more and become more flexible in their language use. However, few studies examine the link between teaching grammar and the skills needed for higher education in the Philippines in the 21st century. Addressing this gap can help curriculum designers and teachers determine how to teach grammar in ways that align with the needs of global conversation and language development as a whole.

Lexical Competence

Lexical competence is a part of communicative competence that refers to a person's ability to understand, recall, and use words appropriately in different situations. It involves more than just knowing the meaning of a word—it also includes understanding its form, common pairings (collocations), and correct use in context. According to López-Pérez et al. (2023), lexical competence may be viewed as a premise of effective communication because the extent of vocabulary knowledge, the number of words known by the learner, and comprehension, the level of language proficiency among the learner, are what determine the degree of proficiency within the whole language. Zare et al. (2021) further explain that lexical richness directly affects fluency and communicative accuracy; thus, vocabulary development is a core aspect of academic and professional success.

Lexical competence has three dimensions: breadth, depth, and use. The breadth refers to the number of words that a learner knows, and depth is the level of information that the learner has of the form, meaning, and usage of a word (Shandra, 2021). In turn, use implies the possibility of applying lexical knowledge in real communicative situations through collocations and idiomatic expressions, as well as the context-specific use of words (Kaur, 2020; Khan & Anjum, 2023). All these dimensions reflect the cognitive and practical aspects of vocabulary acquisition and align with the 4Cs of 21st-century skills: communication, critical thinking, creativity, and collaboration.

One of them is that the broad and rich vocabulary allows learners to share their multidimensional ideas, remain textually, and collaborate in different settings.

Studies show that various teaching strategies can strengthen vocabulary learning. Nishonova (2024) found that using the English language teaching process can not only help students acquire new vocabulary but also build creativity and awareness of context. Similarly, Kudryashova et al. (2020) found that contextual and meaningful exposure to vocabulary promotes long-term memory compared to frequency-based practices. Shandra (2021) notes that interactive vocabulary activities, metacognition reflection, and semantic mapping are active learning strategies that help learners move up the interlanguage continuum, converting passive knowledge of words into active competence.

Moreover, metacognitive awareness plays a key role in vocabulary learning. According to Kaur (2020), students who reflect on the vocabulary-learning method learn better and retain it longer when they do so consciously. Khan & Anjum (2023) reinforced the importance of lexical competence by showing that cognitive associations and word networks help learners form meaningful connections and use both their first and second languages effectively. These findings highlight the importance of incorporating reflection, creativity, and context in teaching to make instruction learner-centered.

Lexical competence also extends beyond the classroom. In professional communication, vocabulary control is vital. Liu & Dou (2023) found that interpreters with strong lexical skills perform better under cognitive pressure because they can select precise words quickly. Shandra et al. (2022) also showed that context-based vocabulary teaching in the information technology field provides learners with lexical materials essential for their professional development. These studies emphasize that vocabulary acquisition should not be limited to academic purposes.

However, educators should implement the e-strategies to prepare learners for professional and interpersonal communication and to enable them to apply lexical skills in authentic contexts.

The lexical competence showed potential for enhancement with the introduction of technology in vocabulary education. Booton et al. (2023) further observed that mobile-aided language learning devices with augmented reality increase learners' interest and enthusiasm, thereby enhancing vocabulary retention. Moreover, researchers might measure lexical development using quantitative tools such as lexical frequency patterns, lexical sophistication scales, and lexical diversity scales (Zare et al., 2021). These measures help educators understand how learners develop vocabulary knowledge and how it influences their communicative competence.

Lexical ability includes language skills, ways of thinking, and the use of language in communication. It includes how big, rich, and functional languages are in different situations and how they support reading, writing, speaking, and listening. However, few studies have provided empirical evidence linking lexical competence to the 4Cs in Philippine higher education, even though existing research recognizes their pedagogical and technological value. Teachers could follow this activity with vocabulary instruction to promote teamwork, innovation, and critical thinking within the communicative approach. In this regard, lexical study will play a crucial role in educational success and work-related preparation.

Semantics

Semantics is the study of the meaning of words, phrases, and sentences. It is one of the most important parts of language because it shows how people express their thoughts and emotions (John Lyons, 2020). In higher education, semantics helps students use the right words, understand more profound meanings, and think beyond literal language in academic and professional settings (Pineda et al., 2024). Today, studies about semantics are not only theoretical

but also practical, helping improve understanding, thinking, and cultural awareness (Mantasiah Prof, 2020).

Semantics combines logic, thinking, and social experience. Classical ideas, such as those of Ibn Jinni, show that word meanings can change depending on how they are used Muid & Mabrurrosi (2022). Mondal (2022) explains that formal semantics is grounded in logic, while cognitive semantics is grounded in personal experience and thinking patterns. Hussien et al. (2023) say that relationships between words — such as opposites, synonyms, and categories — show how people organize vocabulary in their minds and in society. These ideas show that meaning is created through thinking and communication, not fixed beforehand.

The philosophical roots of semantics still influence studies today. (Kusmanto & Pulungan, 2022) explain how meaning relates to truth and interpretation, while Khitam (2023) shows that being aware of meaning helps with understanding and translation. These studies show that semantics is both a mental and structural process that connects language form, thought, and communication.

Teaching semantics helps connect meaning and structure. (Juanda, 2024) asserts, to construct more connections between vocabulary and communicative practices. Speaking of which, as Li et al. (2022) add that people's preferences and biases affect how they create meaning, so teachers should consider these differences. Ahyar & Yusri Ibrahim (2023) found that semantics and pragmatics work together, intertwined in translation, where meaning depends on both linguistic accuracy and the situation. These studies support a teaching approach that helps students think critically and creatively about meaning and use it in real-life communication.

Outside the classroom, semantics also plays a role in social, emotional, and cultural communication. Papic (2023) says that meaning depends on shared understanding, not just grammar. Pritchard (2022) explains that meaning is created through interaction. Themistocleous

(2024) studies how tone and rhythm express emotion in speech, while Medvid et al. (2023) show that semantic analysis can reveal power and manipulation in politics. These studies show that semantics helps people express meaning, identity, and emotion in society.

Semantics also connects with the 4Cs of learning in the 21st century: collaboration through shared meaning, critical thinking through understanding, communication through expression, and creativity through interpretation. So, semantics is not just a language skill but also a thinking and social skill that helps students understand and communicate across cultures and subjects.

Semantics continues to develop, from its philosophical roots to its modern teaching use. It includes structure, thought, and social meaning, helping people create meaning from words and experiences. However, few studies have connected semantics with 21st-century learning skills in higher education. Future research should examine how teaching semantics can enhance understanding, cultural awareness, and communication in real-life settings.

Morphology

Morphology is the study of word structure and how the most minor units of meaning, called morphemes, combine to form words. It plays a vital role in literacy and language learning by helping learners understand how words carry meaning. Morphological awareness is described as the ability to recognize, break down, and manipulate morphemes to decode and construct new words (Altheneyan, 2025) While early research focused on children's literacy, more recent studies have extended the concept to adult and second-language learners, emphasizing its importance in higher education and multilingual settings.

Researchers have found that morphological awareness comprises two key steps: recognizing and manipulating shapes. Finding prefixes, suffixes, and base words is part of recognition. Manipulation means using this information to make or understand new words. (Liu et

al., 2022) found that studying morphemes makes it easier for students to read difficult material. Similarly, Kristensen et al. (2023) showed that students can infer the meanings of new words when they know how words are put together. These studies show that morphology helps people learn new words and understand their meanings by associating word shapes with their meanings.

Morphological knowledge is also beneficial for teaching. Naser & Gandhi (2022) showed that teaching students to examine word parts makes it easier for them to put words together and understand their meanings, helping them read and write more independently. Activities in the classroom, such as finding affixes, learning about word families, and examining derivations, can help students learn new words and better understand what they read. These results make it clear that explicit morphological instruction should be part of second-language programs for both beginners and college students.

Recent studies have also examined how teaching morphology improves comprehension and vocabulary. Learning morphology helps students not only recognize words but also understand them better by connecting words with their meanings. Giazitzidou et al. (2025) report that practicing morpheme use improves students' reading and spelling. They also found that morphological awareness supports vocabulary knowledge and reading comprehension. These studies show that teaching morphology through word-building, affix recognition, and compound-word analysis can help learners improve their vocabulary and understanding.

Cross-linguistic studies show that grammar differs in different languages, which affects how people learn and use it. According to Tafese (2022) Oromo-speaking children have special problems learning inflectional grammar, which slows down their reading and writing growth. Mizan et al. (2023) also showed that knowing Arabic morphemes improves both communication and understanding. Shigini & Mapunda (2023) show that the way the Kemunasukuma people's personal names are put together shows how they see their culture and traditions. These instances

show that morphology is both universal and culturally specific, with different effects across social and linguistic settings.

Additionally, technology has made it easier to learn about and use morphology in more situations. According to Muthee et al. (2022) morphological analysis supports natural language processing (NLP) in artificial intelligence by identifying root words that help computers better understand text. Bollmann and Bollmann & Søgaard (2021) said that machine learning systems struggle with languages with high grammatical richness, suggesting that we need better computer models. In education, digital tools such as live word analysis and mobile apps make learning morphology more fun and easier for students.

To sum up, the growing body of studies on morphology shows that it is used for a lot more than just teaching early reading. Morphological awareness is an important part of language, education, and technology that helps people learn new words, read and write in multiple languages, and use language correctly. Including it in multilingual and tertiary education connects theory and practice, giving students the metalinguistic awareness they need for school, work, and digital communication. Future studies should examine how morphological training and technology-based tools can work together to help students of all kinds understand what they read and learn new words.

Orthography

Orthography is the study of the rules and forms of written language. It is the basis for learning to read, spell correctly, and read quickly. Readers can bridge the gap between spoken and written language by connecting pictures and sounds and understanding their meanings (Chrabaszcz et al., 2023). Most research on literacy development has focused on children. However, new research has recently turned the attention to adults, especially those who speak

English as a foreign language (EFL). Orthography is now an essential part of how well college students decode, understand, and write academic works.

Several studies have shown that knowing a lot about orthography makes reading faster and easier to understand. Johnels et al. (2024) found that there are two types of orthographic knowledge: general and word-specific. They also found that both types of knowledge can predict how well a student will spell from elementary school through high school. At the college level, this result directly affects how English as a Foreign Language (EFL) is taught. Similarly, Zarić & Nagler (2021) claim that knowing how to read words and sentences correctly is a good indicator of how well you will understand what you read. They highlight the importance of diagnostic tests to find orthographic problems in adult learners. These findings show that reading and writing correctly with letters and words is important at all grade levels, especially for students learning to spell English words correctly.

Orthographic processing requires visual attention, pattern identification, and letter–sound relationship. Wong et al. (2023) found that explicit orthographic structure training increases visual discrimination and character processing in young Chinese readers, improving comprehension. (K. H. Huang & Chang, 2021) suggest clear orthographic teaching in early reading development, which can be used in adult and EFL literacy training. Since orthographic skills affect decoding efficiency and spelling correctness, Chen et al. (2020) provide multidimensional evaluation techniques to assess their complexity. These findings support teaching orthographic awareness, which helps students notice letter patterns, word boundaries, and sound–symbol relationships.

From the psycholinguistics perspective, there is still a strong link between understanding written language, paying attention to what you see, and moving between languages. Xu et al. (2021) found that children who have difficulty reading often have problems with orthography. This supports the idea that being aware of orthography can help kids visualize what they are reading.

Tibi et al. (2021) looked at Arabic spelling and found that the way Arabic looks has a big effect on how well you can read it. Ginestet et al. (2020) found that attention and eye movement patterns affect how well we recognize words. This suggests that attention-based techniques can help with learning letters and their sounds. Cross-linguistic studies also show that bilingual people use mental tools they already have to deal with different writing systems. Simone, Williams et al. (2021) argue that the orthographic depth of bilingualism facilitates language transmission in a productive way, thereby improving reading flexibility and comprehension.

In higher education, knowing how to write letters correctly helps with more than just reading and writing. It also helps with writing assignments and expanding your language. Lally & Rastle (2022) found that students who could see letters more clearly were able to read and copy text more accurately. McMurray (2020) supports a balanced way of teaching that includes phonological, orthographic, and morphological factors to improve reading and writing skills. In English as a foreign language (EFL) settings, written materials that offer orthographic feedback help students improve their spelling and reading fluency through computer tools that help students see words. These studies show how important it is to teach spelling in higher education so that students who speak more than one language can improve their advanced reading and writing skills.

In summary, orthographic knowledge is a difficult idea that involves understanding what you see, hear, and understand. Orthographic awareness helps with reading speed, spelling, and understanding what you read, no matter what language you speak or what amount of education you have. Research on children's literacy set the foundation for orthographic training, and recent research shows that it is still useful for adult EFL learners. To help students from a wide range of academic fields and language backgrounds, future study should look into adaptive orthographic training tools, technology-driven visual recognition systems, and cross-linguistic models.

The 4Cs Framework

The 4Cs Framework—Creativity, Critical Thinking, Communication, and Collaboration—is widely regarded as a foundational model for 21st-century education. It represents the essential competencies learners must develop to thrive in a complex and technologically driven society. The framework is primarily attributed to the Partnership for 21st Century Learning (P21) and supported by the Assessment and Teaching of 21st Century Skills (ATC21S) initiative, both of which emphasized adaptability, innovation, and interconnected learning environments as drivers of educational reform. Recent scholarship extends these earlier ideas. Thornhill-Miller et al. (2023) argue that the 4Cs, when systematically reformed, can better align teaching practices with societal and labor market changes. Similarly, De Foo et al. (2021) highlight that the integration of the 4Cs reflects institutional commitment to cognitive flexibility, empathy, and interdisciplinary collaboration. Collectively, recent studies suggest that the 4Cs transcend classroom competencies and serve as key components of lifelong learning and global citizenship.

The most appropriate way of operationalizing the 4Cs in the classroom is by curriculum integration and inquiry-based teaching that involves the students in real-life problem-solving. Izzah & Wulandari (2023) demonstrate how the 4Cs can be designed with the help of Content and Language Integrated Learning (CLIL), where language development is incorporated into the subject of instruction with the purpose of cultivating not only communication and critical thinking skills. This is applied by Siswadi et al. (2024) to the training and education of the professions, and they find that the introduction of the 4Cs to the nursing schools evokes motivation, engagement, and active collaboration in service-learning and volunteer services. These findings confirm that the learners can be equipped with cognitive and interpersonal skills if the 4Cs are well-considered in the curricula. Further, Pardede (2020) also introduces the 4Cs into the framework of leadership

learning, where they introduce it as a model that integrates personal learning and collaborative and reflective learning outcomes, i.e., as character, competency, connection, and culture.

The 4Cs practices and measurement have been re-conceptualized by digital transformation. Technology-based instruction enables both creative thinking and collaborative work with the help of a virtual platform, online discussions, and simulation in the form of a project. Due to the effects of digital tools that facilitate creative thinking and collaborative assessment, as Thornhill-Miller et al. (2023) note, it is now possible to carry out this in a way that was not previously feasible in a conventional classroom. According to Valentijn et al. (2024), technology also increases the level of interdisciplinary interaction, which can result in learners not being limited by the constraints of disciplines or geographical areas. To the same extent, Malik et al. (2020) reveal that digital cooperation contributes to critical thinking and innovativeness of complex systems in mobility and sustainability education- a fact that can be applied to digital learning environments. So, technology is one of the means of the 4Cs; consequently, interaction, creativity, and reflective interactions among learners of various modalities are amplified.

The 4Cs assessment has been a longstanding issue since the traditional types of assessment are less sensitive to the third and fourth order thinking and collaboration. The VALUE rubrics (Valid Assessment of Learning in Undergraduate Education), Torrance Tests of Creative Thinking (TTCT), and California Critical Thinking Skills Test (CCTST) have been suggested as multidimensional tests that are consistent with performance-based results. Both De Foo et al. (2021b) and Do et al. (2023) have advocated the concept of the holistic and integrative method of evaluation that must be directed at the indicators of the individual and group level performance, and integrative systems that must offer the links between the 4Cs and the development of personal leadership and identity. Educational researchers suggest the combination of authentic

assessment in the form of digital portfolios, peer review, and collaborative projects to determine the dynamic interactions of the 4Cs in a practice-based learning process.

Altogether, the 4Cs Framework remains a cornerstone of modern educational reform. It produces flexible, creative, and entrepreneurial learners who are equipped to meet the demands of a global society. While applicable across industries, its most significant impact is in transforming classroom learning by nurturing creativity and critical thinking through inquiry, enhancing communication and technological proficiency, and fostering collaboration through problem-solving. Looking ahead, the 4Cs will continue to inform research, policy, and curriculum design, ensuring that learners are prepared to thrive in a rapidly changing and interconnected world.

Critical Thinking

Critical thinking (CT) is a high-order thinking skill, and it involves the capacity to analyze, evaluate, and synthesize information in order to make good decisions. It provides the foundation of problem-solving, creativity, and reflective judgment at school. Ali et al. (2021) and Hidayatullah et al. (2021) argue that critical thinking is not just the ability to memorize, but also the ability to reason, assess arguments, and make decisions in any sphere. Although teachers are positive about CT,Ali et al. (2021) prove that not all teachers possess comprehensive conceptual understanding and instructional methodology that could enable them to effectively apply it in classroom practice. The latter refers to the urgency of the need for professional growth, which will enable the teachers to transform their knowledge regarding CT into a systematic tool of teaching.

As recent studies demonstrate, the following teaching interventions in the development of CT are effective: problem-based learning (PBL), simulation, and metacognitive training. Hidayatullah et al. (2021) found that the students engaged in the use of PhET simulation-based learning registered improved skills in CT compared to other students in the classroom, particularly

in the analysis, inference, and evaluation skills. No less significant, problem-based learning (PBL) promotes critical thinking and reflection, which Sari & Aminatun (2021) observed in their learning environment on environmental science.

According to Chana (2022), it is also true that more competent students in CT apply deep-learning methods as compared to surface-content strategies, which means that CT, as the pure form, is dictating how students construct and internalise knowledge. In a complementary way, metacognitive interventions, such as those used in a study by, can demonstrate that self-monitoring of thinking leads to a deeper understanding of complex concepts. Sahiruddin et al. (2022) state that reading is predictive of CT proficiency, meaning intensive cognitive activities that take place due to literacy enhancement through thought processes and problem-solving.

The CT-oriented instruction results are always linked with the academic performance and learning transfer. Ali & Awan (2021) State that undergraduate science majors with the highest levels of achievements in analytical and evaluative thinking had higher academic performances, particularly in physics. This finding supports the assumption that the relationship between acquisition and application of knowledge is based on CT. Heim et al. (2022) state that undergraduate science majors with the highest levels of achievements in analytical and evaluative thinking had higher academic performances, particularly in physics. This finding supports the assumption that the relationship between acquisition and application of knowledge is based on CT.

One of the biggest challenges that has been realized in literature is the assessment of CT. Both Heim et al. (2022) and Ayu et al. (2020) mention that there are no standard structures to evaluate CT, and the majority of existing tools are not consistent and aligned with the higher levels of learning outcomes. To surmount this, researchers propose applying multidimensional rubrics and measures that are validated, such as the California Critical Thinking Skills Test (CCTST) and the

VALUE rubrics that assess reasoning, interpretation, and problem-solving. Assessment models should measure the process and outcome of thinking and not the performance level of individual students.

Despite this growing interest throughout the world, little Philippine research has been conducted on the tertiary level CT pedagogy and assessment. Other areas are not exceptions, as Jafari et al. (2020) explain, by indicating that the level of CT among university students is low in places where the curriculum is not specifically structured to teach CT. This, in the Philippine case, implies that curricular redesigning, research-based interventions, and assessment reforms that are aligned with higher-order thinking outcomes are in need. The solution to these gaps would close the gap between theory and practice in the classroom to help educators build CT as an academic skill and a life skill.

All in all, critical thinking is among the most important features of educational change and cognitive development. Such types of interventions as PBL, simulations, and metacognitive training are always evidence-based and assist in the development of analytical reasoning and learning transfer. However, the growth is hampered by definition continuity, incomplete assessment tools, and a paucity of studies on the area. It then follows that there should be one model that establishes pedagogically, outcomes, and assessment of CT, such that the learners are taught to think critically and behave rationally in a more complicated academic and professional environment.

Communication

Communication is a broad and evolving field that includes sociology, education, marketing, and technology (Littlejohn et al., 2021). Both students and professionals need good communication skills to work together, solve problems, and connect with other people in their communities. Effective communication also fosters inclusion and learning by facilitating the

exchange of ideas and experiences in both local and global contexts (Afridah & Lubis, 2024). Recent research indicates that communication is now pivotal to professional achievement, social advancement, and intercultural comprehension in the contemporary digital landscape (Aquino et al., 2024).

In both school and employment, communication involves more than just sharing information. It fosters trust, empathy, and significant connections. Masyhuri et al. (2024) stressed that communication based on local values and culture makes the relationships between school and community stronger. Lapo (2023) also said that teachers are links between schools and communities that encourage respect and collaboration. This research indicates that communication in higher education ought to transcend formal dialogue. It should help people build real relationships that make them want to learn, work together, and be active in their communities. Communication can help develop community and a shared goal when used in this manner.

Digital tools and social media have reshaped how people interact in both personal and professional settings. As Khaeriyah et al. (2023) explained that social media has transformed traditional one-way communication into a two-way, interactive exchange where people collaborate and share ideas openly. Observed that many organizations now use digital platforms to engage audiences and strengthen their reputation. However, Liu et al. (2022) warned that digital communication can lead to distractions and lower productivity when used without discipline. To address this, both students and professionals need strong digital literacy skills to use technology responsibly and effectively.

In a globalized society, intercultural communication has become increasingly important. Yang (2024) pointed out that while online slang and informal expressions make conversations more natural, they can also cause misunderstanding. Dang (2024) argued that digital platforms can

act as spaces for cultural exchange, helping people build global awareness and intercultural understanding. These findings show that digital communication can connect individuals from diverse backgrounds and bridge cultural gaps. Working in multicultural teams or diverse classrooms helps students develop intercultural competence—an essential skill for managing diversity, language barriers, and ethical communication in both digital and real-world contexts.

In today's world, communication is more than just public relations and business. It forms thoughts, builds trust, and helps make social changes. Balonas (2021) claimed that strategic communication should not only help organizations reach their goals but also help communities grow. Pieczka (2020) supports flexible ways of communicating to get people involved in politics and social issues. Setiawan et al. (2021) said that the COVID-19 outbreak made people more reliant on digital platforms, which shows the need for systems that keep people connected. Ramírez et al. (2023) added that being honest and open when talking about corporate social responsibility (CSR) is very important because it builds confidence and trust with the public. All of these studies show that communication today needs to be honest, ethical, and adaptable.

Overall, communication has become a key driver of learning, collaboration, and innovation across disciplines. Strong communication fosters innovation, creativity, and ethical responsibility. Students and professionals must be able to express ideas clearly—whether through speaking, writing, or digital media—while remaining aware of their purpose and audience. As the world becomes increasingly digital and interconnected, communication must remain inclusive, adaptable, and forward-thinking to build meaningful relationships and inspire positive social transformation (Kaur, 2020).

Collaboration

Collaboration is a concept that has gained popularity across many domains due to the growing interdependence between workplaces and schools (Francis et al., 2024). Teamwork helps

individuals develop and refine problem-solving skills and contributes to productivity in both academic and professional settings. In higher education, team-building is an effective strategy for preparing students to become cooperative workers in technologically advanced employment environments. As institutions and industries evolve, it's crucial to understand how collaboration can function effectively as a mechanism that produces innovation, inclusivity, and sustainable outcomes (X. Wang et al., 2024).

Building on this growing recognition of collaboration's importance, recent studies in tertiary education have explored how educators can intentionally design teamwork environments to cultivate creativity, problem-solving, and professional readiness among students. Teamwork in tertiary education facilitates an academic education and professional development. Klooker & Hölzle (2023) state that fostering teamwork and creativity among students requires creating a collaborative innovation environment. In their opinion, educators and institutions may develop social and physical spaces to promote active service and involvement in addressing problems within a group. On the same note, Arinaitwe (2021) discusses institutional cooperation between vocational teacher training and industry, as well as mentoring and acting on research, as means of integrating theoretical education with practical skills in the working environment. These examples show that teamwork in higher education inculcates critical thinking and flexibility, which are fundamental to school and career success.

Extending beyond the context of higher education, insights from other professional domains—particularly health care—offer valuable lessons on how professionals can enhance collaboration. The experience of other fields and health care, in particular, can contribute something valuable to learning how to establish teamwork and collaboration. Abbasi et al. (2022) focus on dental professionals' conflict management and conclude that effective communication affects group performance and outcomes. Moreover, Kida et al. (2024), found that patient safety

climates are bolstered by collaborative leadership and social capital, which are also essential for considering the contribution of teamwork to quality and safety assurance. In turn, Hossny & Sabra (2021) suggest that workplace incivility compromises cooperation between physicians and nurses, demonstrating that respect and civility are essential for successful collaboration. Jafari et al. (2020) add that interprofessional education reduces barriers to cooperation and improves the delivery of care. Based on these insights, schools can use interprofessional methods that encourage students to collaborate, take responsibility for their actions, and become more skilled professionals. The attitude of collaboration and adaptability is becoming more common as digital technology changes how people work and learn together.

In this context, technology has changed how collaboration occurs in both physical and virtual environments. Mirbabaie et al. (2021) note the potential of virtual assistants (VAs) to enhance communication and coordination within a team, especially in remote or hybrid settings where sharing information is particularly challenging. Chantziaras et al. (2021) discuss the use of collaborative robots (cobots) in industrial environments and explain how the relationship between people and machines can assist workers, making them more productive and creative as they perform more complex tasks. In the context of higher education, the identified insights point to the growing importance of digital collaboration tools, virtual exchange, and co-creation platforms that enable students to collaborate in teams without traditional boundaries.

As collaboration becomes more complex and technology-driven, it is essential to monitor its effectiveness continuously. The evaluation of the collaboration needs to be methodical, with means of capturing the process and the results. Hossain et al. (2021) review the challenges of accommodating mentally challenged employees in the workplace and emphasize that without stakeholder cooperation, success is unattainable. In the same way, Campoy-Cubillo & Jiménez Estrada (2023) develop a model for evaluating collaborative work in green manufacturing, in which

dynamic models serve as the foundation for measuring the efficiency and sustainability of the team effort. To ensure collaborative learning is inclusive, equitable, and practical, peer assessment, teamwork rubrics, and reflective evaluations can also be applied in an academic context.

Overall, the existing literature demonstrates that collaboration is essential across education, healthcare, and industry. In the event of successful collaboration, there must be favorable environments, ethical assessment practices, interprofessional learning, and digital tools. Collaborative competence, due to constant development, for students involves communication, managing diversity, and using technology through the joint creation of knowledge. The establishment of teamwork will continue to take center stage in innovation, inclusion, and long-term professional success due to the constant dynamics of workplaces and institutions.

Creativity

Creativity in education is already entering the very heart of the new research activities as teachers and scientists begin to become increasingly more interested in the importance of creativity in developing problem-solving skills, innovation, and cognitive growth (Mangubat et al., 2025). Creativity, which may be defined as the ability to generate original and useful ideas or solutions can be applied in student learning outcomes in different fields, and mathematics and science in particular (G. Baptista, 2025). The publications of modern scientists explore various forms of teaching, formative evaluation, and placement conditions that encourage creative thinking, articulating the possibilities and barriers to developing creativity in organized learning environments.(Panergayo & Pelgone, 2023).

Pedagogical strategies are quite critical in inculcating creativity, whereby the students are involved in inquiry, collaboration, and reflection. The philosophy of Problem-Based Learning (PBL) in facilitating the cultivation of creative thinking among students is also supported by Permata et al. (2022), who encourage the emergence of different solutions to environmental issues in the real

world. Similarly, Fatmawati et al. (2022) cannot determine that the Creative Problem-Solving (CPS) model encourages divergent thinking, suggesting that the instructional design of a problem directly affects creative outcomes. Khalid et al. (2020) also add that creative problem solving in mathematics is correlated with enhancing analytical skill, yet developing broader cognitive skills, not restricted to academic achievement. Jaenudin et al. (2020) verify this by showing that Project-Based Learning (PjBL) provides the balance between theory and practicality that allows students to apply creative knowledge to practical issues. All this literature proves the idea that creativity is flourishing where the teaching practices are based on inquiry, teamwork, and experiential learning.]

A crucial, albeit complex, part of the educational design is also the evaluation of creativity. Rahayuningsih et al. (2021),found that the authors of the current research emphasize the importance of the use of open-ended problems to solve, which would assess the ability of students to find and choose different paths to solutions- a strategy that is consistent with the Torrance Tests of Creative Thinking (TTCT) and other rubrics that would have assessed originality, fluency, and flexibility. Aliffia & Gusti Ketut Agung Ulupui (2023) expand on the debate by relating creativity to emotional intelligence and showing that students with balance in emotions are more flexible in thinking and possess a higher level of performance in solving problems. A combination of these studies suggests that assessment systems should include both cognitive and affective aspects of creativity to capture the full range of student innovation.

Despite the emphasis on creative learning, it is often suppressed by institutional constraints. Zhan et al. (2024) present the evidence suggesting that the inflexibility of curricula and the normative type of educational assessment system could suppress the creative powers of students, even within the system of problem-based learning. Also, Abuhadeeda et al. (2024) refer to the significance of social dynamics because the authors claim that though collaborative learning

settings are linked to creativity, their performance depends on the group's co-position, diversity, and learners' engagement. These findings imply that to encourage creativity, both effective pedagogical models and less restrictive institutional regimes that promote experimentation, inclusiveness, and flexibility are required.

Technology has provided new opportunities for fostering innovation and more access to collaborative innovation. According to Ndlela & Barnes (2024), the example of creative problem-solving can be offered, in which digital tools (e.g., creative software, online platforms, virtual collaboration environment) may be utilized to broaden the perspectives and enable new forms of interaction. This aligns with the teacher's demands to integrate digital literacy into creative pedagogy, where students should not just use technology to consume information but should design, explore, and be creative. Technology-enhanced creativity could be empowering because learning rooms are increasingly digitalized, allowing learners to become more critical in their thinking and act in finding solutions to real-world problems.

The growing number of publications signifies the leading role of creativity in modern education. Good instruction, which is problem-solving, critical, and reflective in nature, promotes creative development; creative assessment, flexible curriculum, and technology maintain it. However, these institutional barriers, which cannot be overcome immediately, remind educators that they need to re-create a learning environment where originality and exploration are appreciated. As education is dynamic, a creative strategy is essential for providing learners with greater adaptability to innovate and survive in an ever-changing world.

1.3 Theoretical Framework

This study is anchored on Dell Hymes' Theory of Communicative Competence (Hymes, 1972), which challenged the structuralist view that linguistic knowledge—particularly grammar—

was the primary basis of language proficiency. Hymes (1972) argued that effective communication requires not only mastery of grammatical rules but also the ability to use language appropriately across social contexts. He emphasized that communicative competence involves understanding when, how, and to whom language should be used, reflecting both linguistic form and sociocultural context.

Building on Hymes' sociolinguistic foundation, Canale & Swain (1980) proposed a more systematic model consisting of four interrelated components. Grammatical competence refers to knowledge of vocabulary, syntax, morphology, and phonology, which allows learners to construct meaningful and accurate sentences. Sociolinguistic competence involves understanding social norms, cultural conventions, and context-appropriate language use. Discourse competence enables individuals to organize ideas cohesively in both spoken and written communication, while strategic competence pertains to the use of verbal and non-verbal strategies to repair breakdowns and enhance message clarity. Within this model, linguistic competence remains a core element, as it underpins learners' ability to interpret messages, produce coherent discourse, and participate successfully in communicative interactions.

To extend the discussion from linguistic ability to broader educational competencies, this study also draws on the P21 Framework for 21st Century Learning, developed by the Partnership for 21st Century Learning P21 (2019) This framework highlights the importance of the 4Cs—Critical Thinking, Communication, Collaboration, and Creativity—as essential skills required for success in contemporary academic and real-world environments. The P21 Framework positions communication not merely as the exchange of information, but as the clear articulation of ideas, active listening, and engagement in meaningful dialogue. It likewise views critical thinking as the ability to analyze information and solve problems, collaboration as coordinated interaction toward shared goals, and creativity as the generation of original and innovative ideas. These

competencies collectively reflect the cognitive, social, and linguistic demands of 21st-century learning.

Integrating these perspectives, this study conceptualizes linguistic competence as the foundational resource that supports both communicative competence and the development of the 4Cs. Strong linguistic competence enhances learners' ability to think critically by enabling them to comprehend complex texts, evaluate information, and express reasoned judgments. It likewise contributes to effective collaboration through clear interaction and shared understanding, and it enriches creativity through flexible and innovative use of language. Communication—the second major construct in both Hymes' theory and the P21 Framework—serves as the bridge connecting linguistic skills to 21st-century competencies.

Thus, the Theory of Communicative Competence (Canale & Swain, 1980; Hymes, 1972) and the Partnership for 21st Century Learning (P21) (2019) jointly provide a comprehensive foundation for the present study. Together, they explain how linguistic competence shapes students' communicative abilities and why these abilities are essential for achieving the critical thinking, collaboration, communication, and creativity skills demanded in modern education. These theoretical lenses justify examining the relationship between students' linguistic competence and their perceived 4Cs, highlighting language proficiency as both a communicative necessity and a catalyst for broader cognitive and social development.

1.4 Systematic Diagram

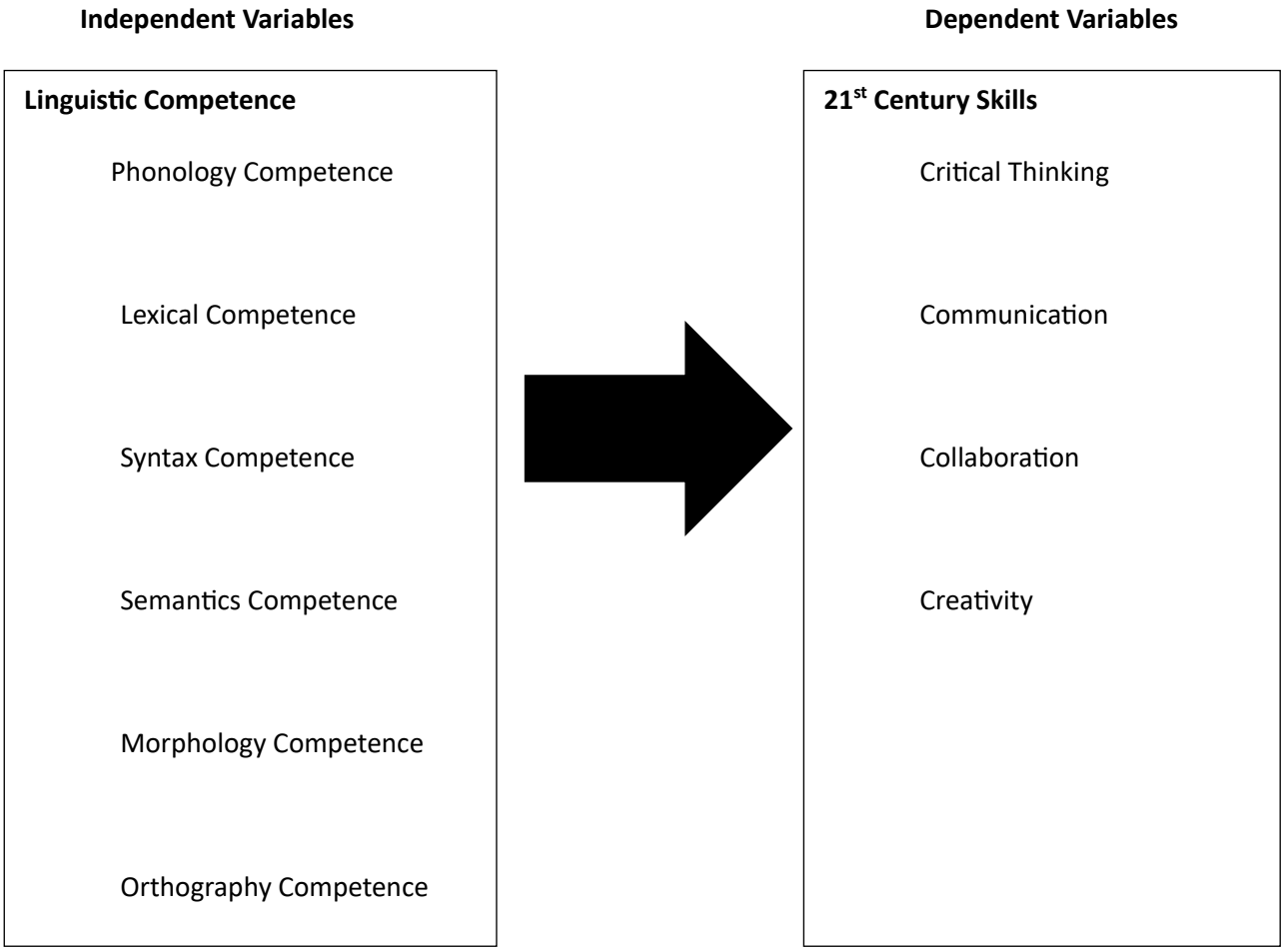


Figure 1. The schematic diagram shows the correlation between the independent and dependent variables of the sample.

1.5 Statement of the Problem/ Objectives of the Study

The study will aim to determine the relationship between 21st-century skills and linguistic competence among first-year students in the CTE and CAS departments at Saint Michael College of Caraga.

Specifically, the research seeks to address the following questions:

1. What is the level of college students' linguistic competence in terms of:

1.1 phonology

1.2 lexical,

1.3 syntax,

1.4 semantics,

1.5 morphology; and

1.6 orthography?

2. What is the level of 21st-century skills among college students in terms of:

2.1 communication

2.2 collaboration

2.3 critical thinking; and

2.4 creativity?

3. Is there a significant relationship between students' linguistic competence and their 21st-century skills?

1.6 Hypotheses

This study will test the following null and alternative hypotheses:

Null Hypothesis (H_0):

There is no significant relationship between students' linguistic competence and their 21st-century competencies.

Alternative Hypothesis (H_1):

There is a significant relationship between students' linguistic competence and their 21st-century competencies.

1.7 Scope and Limitation of the Study

This study examines the relationship between linguistic competence and 21st-century competencies—specifically communication, collaboration, critical thinking, and creativity—among first-year students of the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS) at Saint Michael College of Caraga during the First Semester of School Year 2025–2026.

The scope of linguistic competence in this study is limited to six components: phonology, lexical competence, syntax, semantics, morphology, and orthography. These areas form the basis of the researcher-made linguistic competence test administered to the participants. The study does not include other language domains such as discourse competence, pragmatic competence, or conversational English. The study focuses solely on academic English skills.

For the 21st-century competencies, the study is confined to the four core skills identified in the P21 Framework: communication, collaboration, critical thinking, and creativity. These were measured using an adapted questionnaire developed by Varona (2020) Other 21st-century skills, such as digital literacy, media literacy, leadership, and global awareness, are not included.

The participants of this study are limited to first-year students enrolled in selected programs under CTE and CAS. Only the following programs are included: BSED-English, BSED-Science, BSED-Math, BSED-Social Studies, BEEd, BPED, BTVTED, and BHumServ. Students from other colleges, such as Business Administration, Criminology, Computer Studies, and Maritime Education, are excluded. Only first-year students were chosen because they share common foundational language and general education courses; thus, they provide a consistent baseline for assessing linguistic competence.

The study is further limited by its use of purposive sampling, which means the results cannot be generalized to the entire SMCC population. In addition, the use of self-report questionnaires to measure the 4Cs may introduce response bias, as students may respond based on personal perceptions or social desirability. The linguistic competence test, although validated by experts, is researcher-developed and may not fully capture students' actual linguistic proficiency.

The study also does not account for external factors that may influence both linguistic competence and 21st-century skills, such as students' previous schooling, socioeconomic background, exposure to English outside the classroom, teacher effectiveness, or family environment. These variables are outside the scope of the present study.

Lastly, the research uses a descriptive–correlational design and therefore cannot establish cause-and-effect relationships. It can only identify whether a significant relationship exists between linguistic competence and the 4Cs.

1.8 Definition of Terms

To ensure conceptual precision and clarity in the use of these terms, which are operationally defined for the purposes of this study, the following terms will be used:

21st Century Competencies (4Cs). This refers to the core skills of critical thinking, communication, collaboration, and creativity, which are exemplified by the first-year students of the College of Teacher Education, Arts and Sciences at Saint Michael College of Caraga. These skills will be analyzed in relation to their achievement and participation in classroom tasks.

Critical Thinking. This indicates the respondents' ability to analyze, evaluate, and synthesize information in a systematic way for the purpose of making sound judgments.

Communication. This pertains to the ability of the respondents to express their ideas as well as communicate information effectively, both orally and written.

Collaboration. This refers to the respondents' ability to work together with other learners in the class. This is represented through group projects, cooperative learning activities, and other similar group work undertaken in the classroom.

Creativity. This denotes the capacity of the respondents to be creative, innovative, and imaginative in producing academic outputs and presentations.

Linguistic Competence. This captures the level of respondents' proficiency in using English in oral and written academic activities correctly and appropriately.

Phonology. This refers to the respondents' ability to recognize, distinguish, and produce the correct sounds of English, including pronunciation, stress, rhythm, and intonation.

Lexical Competence. This is associated with the vocabulary knowledge of the respondents and their ability to use words accurately and appropriately in their essays, discussions, and examinations.

Syntax. This refers to the respondents' command of word order and sentence structure in the construction of grammatically sound, contextually appropriate oral and written communication.

Semantics. This points to the respondents' awareness and accurate usage of word meanings, sentence and text understandings, and demonstration of such understandings in reading and in classroom conversation.

Morphology. This refers to the respondents' ability to accurately encode and decode words using knowledge of prefixes, suffixes, and root words in a variety of academic tasks.

Orthography. This captures whether the respondents' written work follows the proper conventions of spelling, capitalization, and punctuation.

CHAPTER 2

METHODOLOGY

2.1 Research Design

The study will employ a descriptive–correlational quantitative research design to examine the relationship between linguistic competence and 21st-century competencies—specifically communication, collaboration, critical thinking, and creativity—among first-year students of Saint Michael College of Caraga. The study will use a descriptive design to determine the students' levels of linguistic competence and their perceived 4Cs. In contrast, the correlational component will assess whether a significant relationship exists between these variables.

The study will adopt a quantitative approach to enable the objective measurement, analysis, and interpretation of numerical data. Descriptive statistics, such as mean, standard deviation, and frequency distributions, will be used to summarize the data. In contrast, inferential statistics, particularly the Pearson Product-Moment Correlation Coefficient (r) and, when necessary, Spearman's rho, will be used to test the degree and direction of association between the independent and dependent variables. This design is most appropriate because it will not involve manipulating variables but instead examine existing relationships, enabling the researchers to determine whether linguistic competence is significantly associated with students' development of 21st-century skills.

2.2 Research Materials and Methods

Two main research instruments will be used in this study: a researcher-made linguistic competence test and an adapted 21st-century competencies questionnaire.

The linguistic competence test will be developed by the researchers to measure students' proficiency in six core areas of English: phonological, lexical, syntactic, semantic, morphological, and orthographic competence. The instrument will consist of 120 items in both multiple-choice and short-response formats. Before its use, the test will be evaluated by three English language experts to ensure clarity, relevance, and alignment with the study objectives. A pilot test will be conducted with 20 students who are not part of the main sample to determine reliability. Internal consistency will be measured using Cronbach's alpha, with a coefficient of 0.80 or higher considered acceptable. (George & Mallery, 2016)

The 21st-century competencies questionnaire, adapted from (Varona, 2020) will assess students' perceived levels of communication, collaboration, critical thinking, and creativity. Each competency will include five indicators, resulting in 20 items rated on a four-point Likert scale ranging from 1 (Strongly Disagree) to 4 (Strongly Agree). The adapted instrument will undergo expert validation to confirm contextual appropriateness and will be subjected to factor analysis to establish construct validity. Reliability will likewise be measured using Cronbach's alpha, with acceptable values ranging from 0.85 to 0.95.

Prior to data collection, the researchers will secure permission from the deans of the concerned departments. The instruments will be administered in a controlled classroom setting to ensure standardized testing conditions. Participants will be given sufficient time to complete both the linguistic competence test and the questionnaire. The researchers will personally collect all responses to ensure the completeness and accuracy of the data.

2.3 Research Environment, Population, and Sample

The study will be conducted at Saint Michael College of Caraga (SMCC) in Nasipit, Agusan del Norte. SMCC is a private Catholic institution offering academic programs aimed at developing students' competencies in communication, literacy, and 21st-century skills. The research will focus on the College of Teacher Education (CTE) and the College of Arts and Sciences (CAS), which both offer courses related to English proficiency and communication.

The population of the study consists of all first-year students enrolled in CTE and CAS during the First Semester of School Year 2025–2026. Using purposive sampling, the researchers will select 73 first-year students who are enrolled in courses that include English proficiency, communication, or language-related subjects. The sample will include students from the following programs:

- Bachelor of Secondary Education (majors in English, Science, Mathematics, and Social Studies)
- Bachelor of Elementary Education (BEEd)
- Bachelor of Physical Education (BPEd)
- Bachelor of Technical-Vocational Teacher Education (BTVTEd)
- Bachelor of Human Services (BHumServ)

Purposive sampling is used because the study requires participants who are currently exposed to English and communication-related coursework, which is directly aligned with the variables being measured.

Before data collection, the researchers will secure approval from the deans of the concerned departments. Participants will also be informed about the nature and purpose of the study and will be asked to sign an informed consent form, ensuring voluntary participation, confidentiality, and ethical compliance.

2.4 Research Data and Analysis Plan

The data collected from the research instruments will be encoded, organized, and analyzed using the Statistical Package for the Social Sciences (SPSS) version 26. Descriptive statistics, such as mean and standard deviation, will be used to determine the students' levels of linguistic competence and 21st-century skills. Frequency and percentage distributions will be used to summarize the respondents' demographic information and performance levels.

To test the hypotheses, the Pearson Product-Moment Correlation Coefficient (r) will be employed to examine the strength and significance of the relationship between linguistic competence (independent variable) and the four dimensions of 21st-century skills—communication, collaboration, critical thinking, and creativity (dependent variables). The level of significance will be set at $\alpha = 0.05$. A p-value less than or equal to 0.05 will lead to the rejection of the null hypothesis; otherwise, it will be retained.

Before conducting correlation analysis, the assumptions of normality and linearity will be examined using the Shapiro–Wilk test and scatterplot inspection. If the data do not meet the assumptions for parametric testing, the Spearman's rho correlation coefficient will be used as an alternative. To complement the interpretation of the correlations, effect sizes (r) will be reported using Cohen's (1988) guidelines:

- 0.10 = small,
- 0.30 = medium,
- 0.50 = large.

Correlation strength will be described using the following conventional ranges:

- 0.00–0.19 (very weak)
- 0.20–0.39 (weak)
- 0.40–0.59 (moderate)
- 0.60–0.79 (strong)
- 0.80–1.00 (very strong)

Before analysis, all data will be checked for completeness and consistency. Missing or invalid responses will be removed using listwise deletion, provided the proportion of missing data remains minimal. All statistical results will be presented in tables and interpreted in relation to the research objectives and hypotheses.

2. 5 Ethical Standards

The researchers will strictly adhere to established ethical principles to ensure integrity, fairness, transparency, and the protection of all participants throughout the study. The following standards address all requirements related to intellectual property, informed consent, confidentiality, participant safety, data integrity, and responsible use of technology.

A. Protection of Intellectual Property Rights (IPR)

This study does not involve the creation of new technologies, products, or processes requiring patent protection. However, all academic works, research instruments, and instructional materials used are protected under copyright laws. The researchers will duly acknowledge all

authors, including the 21st Century Skills Questionnaire by Varona (2020). No copyrighted or proprietary material will be reproduced, shared, or modified beyond its permitted academic use. All visual symbols, logos, and institutional identifiers used in the study will only appear with proper authorization.

B. Informed Consent

Prior to participation, all respondents will receive a clear explanation of the study's purpose, scope, procedures, expected time commitment, benefits, and minimal risks. They will be informed of their rights, including the right to withdraw at any time, skip questions they are uncomfortable with, and ask clarifying questions. Written informed consent will be secured from all participants before data collection begins. Since the study does not involve animals or clinical human trials, additional institutional clearance for such procedures is not applicable.

C. Data Privacy and Confidentiality.

The researchers will collect only the information necessary for academic analysis and will not ask for sensitive personal details such as ethnicity, political affiliation, or medical history. All respondent identities will be anonymized using codes instead of names. Data will be stored securely in password-protected digital drives accessible only to the research team. Printed materials, if any, will be kept in locked storage. No third party will gain access to raw data unless participants give explicit written consent. Only aggregated and anonymized results will be presented in the final report.

D. Voluntary Participation and Freedom to Withdraw.

Participation in this study is entirely voluntary. Respondents may withdraw at any point without penalty, judgment, or negative academic implications. They may also skip any item they feel uncomfortable answering. No incentive or coercive element will be used to pressure participants. Only those who willingly sign the consent form will take part in the study.

E. Minimization of Harm and Risk Management.

Because the study involves only non-invasive survey questionnaires, no physical, psychological, or social harm is anticipated. Nonetheless, the researchers will remain vigilant in ensuring participant comfort. If any respondent experiences discomfort, distress, or a health concern during data collection, the researchers will immediately pause the procedure for that participant. They will be allowed to rest or discontinue participation without consequence. The researchers will notify the assigned faculty member and, if necessary, assist the participant in seeking support from the school clinic or health services unit. All institutional emergency and health protocols will be strictly followed. The participant's safety, comfort, and autonomy will always take priority over the research process.

F. Beneficence and Contribution to Knowledge.

This research aims to contribute to educational improvement by examining the relationship between linguistic competence and 21st-century skills. The findings may serve as a basis for curriculum development, instructional planning, and learner support programs. Participants involved in the study will have access to the summary results, which summarize general performance and trends of the group. This access will allow students to identify common strengths and weaknesses and gain insight into areas where they can focus their efforts for improvement. Individual responses will remain strictly confidential to protect privacy.

G. Justice and Fair Participant Selection.

The selection of participants will follow transparent, fair, and non-discriminatory criteria. All eligible first-year CTE and CAS students will be invited to participate regardless of gender, socioeconomic status, religion, academic performance, or cultural background. The sampling method will ensure equal opportunity for participation and will not exclude or favor any group without valid academic justification.

H. Data Integrity and Accuracy.

The researchers will employ standardized procedures for data collection, encoding, and statistical analysis to uphold reliability and accuracy. All data will be double-checked to minimize errors. Biases, limitations, or inconsistencies will be openly acknowledged. Under no circumstance will the researchers fabricate, manipulate, or omit data, as academic honesty and integrity remain central to the conduct of the study.

I. Transparency and Honesty in Reporting.

All findings, whether significant or non-significant, will be reported truthfully and objectively. The researchers will not exaggerate results or withhold information relevant to the interpretation of the study. All sources, including copyrighted and patented materials, will be properly cited following APA 7th edition guidelines. Limitations of the study will also be explicitly stated.

J. Use of Patented or Copyrighted Materials.

Although the research does not employ patented technologies, it uses copyrighted academic instruments. For these materials, permission will be requested when required, and proper acknowledgments will be made. The researchers will ensure that the questionnaire by Varona (2020) and any other validated instruments are not distributed, modified, or reproduced outside the approved research context.

K. Ethical Considerations for Animal and Human Trials.

The study does not involve animals or laboratory-based human trials. For human participants, ethical guidelines have been strictly followed, including informed consent, voluntary participation, and minimal risk. Since the data collection involves only self-administered questionnaires, no invasive procedures or hazardous activities are included.

L. Responsible Use of AI and Related Technologies.

AI tools were used responsibly and only for grammar checking, paraphrasing assistance, formatting, and citation organization. AI programs were not used to generate, alter, or manipulate research data or results. All interpretations, conclusions, and written content remain the researchers' original work. By using AI ethically, the researchers ensure the integrity, authenticity, and credibility of the study

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ORGANIZATION	POSITIONS HELD

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ORGANIZATION	POSITIONS HELD

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Elementary : Camagong Elementary School
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ORGANIZATION	POSITIONS HELD

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Elementary : Buenavista Central Elementary School
Secondary : Saint James Highschool INC
Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

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Elementary : Talisay Central Elementary School SPED
Secondary : Saint Michael College of Caraga
Tertiary : N/A

Membership/Affiliations

ORGANIZATION	POSITIONS HELD

APPENDICES

A- Letter of Approval with Certificate of Appearance and Photo

B- Instrument with Informed Consent

21ST CENTURY QUESTIONNAIRE

Name (Optional): _____
Sex: ☐ Male ☐ Female
Age: _____
Course & Year: _____
Department: ☐ CTE ☐ CAS

Level of Critical Thinking Skills

	Strongly Agree	Agree	Disagree	Strongly Disagree
Questionnaire	**4**	**3**	**2**	**1**
I can understand the link connection between ideas.				
I can see the importance of ideas.				
I think of topic/issue in an objective way.				
I think independently.				
I try to look for solutions to problems.				
I evaluate generated information				
I put critical thinking skills to be used in aspects of everyday life.				
I develop my reasoning skills regularly.				
I manifest in the skillful manipulation of my own ideas/interests.				
I can compare ideas.				

Level of Communication Skills

	Strongly Agree	Agree	Disagree	Strongly Disagree
Questionnaire	**4**	**3**	**2**	**1**
I can communicate effectively with others.				
I ask questions if received messages are not clear to me.				
I convey clear meanings.				
I keep communication confidential when needed.				
I am an active listener.				

I believe being skillful in communicating provides freedom to express my thoughts/feelings properly.				
I keep communication confidential when needed.				
I use gestures to help me express my thoughts/feelings to others.				
I see to it that I interpret received messages clearly.				
I respect other people's point of view.				

Level of Collaboration Skills

	Strongly Agree	**Agree**	**Disagree**	**Strongly Disagree**
Questionnaire	**4**	**3**	**2**	**1**
I take time to respond to group problems/needs.				
I facilitate contributions of group members.				
I participate in a class project to attain a common goal.				
I contribute to group assignments.				
I foster a constructive group climate.				
I love to work when it is done by group.				
I believe that teamwork is essential at all times.				
I analyze problems without blaming others.				
I consider each one's weakness.				
I take a leadership role.				

Level of Creativity

	Strongly Agree	**Agree**	**Disagree**	**Strongly Disagree**
Questionnaire	**4**	**3**	**2**	**1**
I consider myself as a creative person.				
I am self-motivated to use my imagination.				
I like to do creative works in class.				
I explore different perspectives.				
I make connections to ideas.				
I enjoy doing creative works.				
I examine things carefully to understand what they mean.				
I organize my ideas well so that my classmates will be able to understand what I suggest.				
I have a particular time during the day when I do most of my creative works.				

I have a particular place where I do most of my creative works.				

LANGUAGE PROFICIENCY TEST

SECTION A. LEXICAL COMPETENCE (Items 1–20)

Focus: vocabulary precision, collocation, register, connotation.

1. (B1) Which word best completes the sentence: “The meeting was very long and honestly quite ____; I almost fell asleep”?

A. exciting B. boring C. loud D. dangerous

2. (B1) Which word best matches the meaning “to make something better”?

A. reduce B. improve C. divide D. pretend

3. (B1) Which option shows the most natural collocation?

A. do a mistake B. make a mistake C. create a mistake D. build a mistake

4. (B1) Which word best completes the academic sentence: “This study will ____ the effects of sleep deprivation on memory”?

A. guess B. copy C. examine D. borrow

5. (B1) Which option most nearly defines “deadline”?

A. a line you cannot cross B. the final version of a story

C. a fixed time when something must be finished D. a threat from your boss

6. (B2) Which option best replaces **really good** in formal written English?

A. fairly okay B. highly effective C. kind of fine D. massively huge

7. (B2) Which word best completes the sentence: “The counselor encouraged the student to seek emotional ____ after the incident”?

A. repair B. healing C. fixing D. glue

8. (B2) Which word best completes the sentence: “The results were ____; they clearly supported our original hypothesis”?

A. confusing B. inconclusive C. predictable D. conclusive

9. (B2) Which phrasal verb best completes the sentence: “We need to ____ the main issues before we can move forward”?

A. break out B. deal with C. run over D. cut off

10. (B2) Which sentence uses an appropriate academic register?

A. The data shows that stress is kind of bad for sleep.

B. The data literally proves stress ruins sleep.

C. The data indicates a strong link between stress and poor sleep.

D. The data is crazy about sleep and stress.

11. (B2) Which sentence shows correct and natural collocation?

A. She did heavy progress this month.

B. She made heavy progress this month.

C. She made significant progress this month.

D. She did a significant progress this month.

12. (B2) Which word best completes the sentence: “The new policy is designed to ____ equal opportunities for all students”?

A. promote B. translate C. confuse D. erase

13. (B2) Which word is closest in meaning to “credible”?

A. believable B. fashionable C. forgettable D. avoidable

14. (B2) What does “tension” most nearly mean in “There was growing tension between the two departments”?

A. excitement and celebration B. physical stretching

C. stress and conflict D. silence and peace

15. (B2) Which word best completes the sentence: “The professor gave a brief ____ of the article before beginning the discussion”?

A. summary B. summarize C. summarized D. summarizing

16. (C1) Which word best fits the nuance: “Her tone during the meeting was noticeably ____”?

A. hostile B. diplomatic C. explosive D. amused

17. (C1) Which word best completes the sentence: "The study reveals a highly ____ relationship between diet and cognitive performance"?

A. random B. accidental C. insignificant D. significant

18. (C1) Which option conveys a formal, objective tone?

A. completely destroyed the other person's argument

B. kind of won the argument

C. presented a compelling counterargument

D. roasted the other person

19. (C1) Which word best completes the sentence: "Many social media campaigns rely on emotional ____ to influence public opinion"?

A. manipulation B. homework C. translation D. recycling

20. (C1) Which word best completes the sentence: "The witness eventually ____ that parts of her original statement were inaccurate"?

A. confessed B. complained C. repeated D. pretended

SECTION B. SYNTACTIC COMPETENCE (Items 21–40)

Focus: grammar accuracy, sentence structure, agreement, subordination, cohesion.

21. (B1) Which sentence is grammatically correct?

A. He do not understand the question.

B. He does not understand the question.

C. He do not understands the question.

D. He did not understands the question.

22. (B1) Which sentence uses the correct verb form?

A. My friends was studying when I called.

B. My friend were studying when I called.

C. My friend was studying when I called.

D. My friends is studying when I called.

23. (B1) Which sentence has the correct word order for adverbs of frequency?

A. Always she is late.

B. She is always late.

C. She always is late.

D. Late always she is.

24. (B1) Which past-tense form correctly completes: "Yesterday we ____ the results of the experiment"?

A. discuss B. discussed C. discussing D. were discuss

25. (B1) Which option is a complete sentence?

A. Because the library was closed.

B. After the meeting.

C. The library was closed for repairs.

D. When I arrived at school.

26. (B2) Which sentence shows correct subject–verb agreement?

A. The number of applicants have increased.

B. The number of applicants has increased.

C. The numbers of applicant has increased.

D. The numbers of applicant have increased.

27. (B2) Which sentence maintains parallel structure?

A. She values honesty, being punctual, and responsibility.

B. She values to be honest, to be punctual, and responsibility.

C. She values honesty, punctuality, and responsibility.

D. She values honesty, to arrive on time, and responsibility.

28. (B2) Which revision best corrects the run-on?

A. The article was difficult to read; it was full of technical terms.

B. The article, was difficult to read, it was full of technical terms.

C. The article was difficult. To read, it was full. Of technical terms.

D. The article was difficult, because full of technical terms it was.

29. (B2) Which sentence uses the passive voice correctly?

A. The experiment was conduct by the assistant.

B. The experiment was conducted by the assistant.

C. The experiment conducted by the assistant.

D. The experiment has conduct by the assistant.

30. (B2) Which sentence places the modifier correctly?

A. Running to catch the bus, the folder fell out of his hands.

B. The folder fell out of his hands running to catch the bus.

C. Running to catch the bus, he dropped the folder.

D. He dropped the folder running, to catch the bus.

31. (B2) Which option maintains logical tense sequence in reported time?

A. She said she will submit the paper tomorrow.

B. She says she would submit the paper tomorrow.

C. She said she would submit the paper the next day.

D. She says she will submitted the paper the next day.

32. (B2) Which sentence is clearest and correctly punctuated with a nonessential clause?

A. The interview, which lasted two hours, it provided valuable insight.

B. The interview, which lasted two hours, provided valuable insight.

C. The interview which lasted two hours, provided valuable insight.

D. The interview which lasted two hours provided valuable insight.

33. (B2) Which verb form correctly completes the second conditional: “If the university ____ more mental health counselors, students would feel better supported”?

A. hire B. hires C. hired D. hiring

34. (B2) Which reported speech option is correct for “I am overwhelmed,” she said?

A. She said that she is overwhelmed.

B. She said that she was overwhelmed.

C. She said that I was overwhelmed.

D. She said that she has overwhelmed.

35. (B2) Which sentence is NOT a run-on?

A. Many students struggle with anxiety it affects their performance.

B. Many students struggle with anxiety, it affects their performance.

C. Many students struggle with anxiety, which affects their performance.

D. Many students struggle with anxiety it, affects their performance.

36. (C1) Which sentence is the most concise and accurate for cause?

A. Due to the fact that the equipment was outdated, productivity decreased.

B. Because the equipment was outdated, productivity decreased.

C. Productivity decreased because of the fact that equipment was outdated.

D. Productivity that decreased was because the equipment was outdated.

37. (C1) Which sentence is best for academic writing and apposition?

A. The participants, most of them first-year students, were experiencing high stress.

B. The participants were first-year students, most of them, and experiencing high stress.

C. The participants, experiencing high stress, most of them first-year students.

D. Experiencing high stress, most of them, the participants were first-year students.

38. (C1) Which option uses subordination to show cause and effect?

A. The server crashed, we lost the data.

B. The server crashed because we lost the data.

C. Because the server crashed, we lost the data.

D. We lost the data, because, the server crashed.

39. (C1) Which sentence has a clear pronoun reference?

A. When the teacher met with the student, she felt calmer.

B. When the teacher met with the student, the student felt calmer.

C. When the teacher met with the student, it felt calmer.

D. When the teacher met with the student, this felt calmer.

40. (C1) Which complex sentence is grammatically correct?

A. Although many disagree, but the policy will remain in place.

B. Although many disagree, the policy will remain in place.

C. Many disagree although, the policy will remain in place.

D. Many disagree although the policy will remain in place.

SECTION C. SEMANTIC COMPETENCE (Items 41–60)

Focus: meaning, inference, figurative language, semantic relations.

41. (B1) What does “break the ice” mean in “To break the ice, the speaker told a short joke”?

A. Make the room colder B. End the event early

C. Start a conversation more comfortably D. Criticize the audience

42. (B1) Which pair consists of antonyms?

A. increase / rise B. urgent / immediate C. deny / admit D. finish / complete

43. (B1) Which sentence shows disappointment?

A. That is exactly what I hoped for!

B. This is not what I expected at all.

C. This is perfect.

D. This is better than I imagined.

44. (B1) Which sentence uses figurative (not literal) meaning?

A. The classroom was freezing.

B. The classroom was 12°C.

C. The classroom had open windows.

D. The classroom was cold because of the air conditioner.

45. (B1) Which sentence suggests possibility rather than certainty?

A. She will arrive at 8:00 p.m.

B. She is arriving at 8:00 p.m.

C. She might arrive at 8:00 p.m.

D. She arrived at 8:00 p.m.

46. (B2) What does “rebuild public trust” most likely mean after a scandal?

A. convince people to forget the past

B. regain credibility and confidence from the public

C. stop answering questions from the press

D. hide the details from reporters

47. (B2) What is implied by “I guess that is one way to solve the problem,” she said, raising an eyebrow?

A. full agreement B. solution is excellent C. skepticism or doubt D. lack of understanding

48. (B2) Which sentence signals contrast?

A. therefore, she passed B. as a result, she passed

C. however, she still failed D. consequently, she passed

49. (B2) Which option shows inference from behavior?

A. He slammed the door and did not speak to anyone for the rest of the day.

B. The door is made of wood.

C. He spoke to everyone.

D. He enjoyed the conversation.

50. (B2) Which sentence contains an idiom?

A. meet at 3 p.m. B. hit the books C. chair is wooden D. She is 20

51. (B2) Which statement best compares register?

A. The patient experienced nausea overnight.

B. The guy felt kind of sick last night.

C. Both A and B mean the same, but A is more formal.

D. Both A and B mean different things.

52. (B2) Which paraphrase best matches “Participation is mandatory”?

A. You can join if you want.

B. Joining is required.

C. You probably should join.

D. It is better not to join.

53. (B2) What is implied by “Her apology sounded rehearsed”?

A. natural and honest B. memorized and not fully sincere

C. shouted in anger D. written on paper

54. (B2) In which sentence is “literally” used correctly?

A. I literally died.

B. I was literally shaking during the earthquake.

C. That joke was literally hilarious.

D. I literally cannot even.

55. (B2) Which word best completes: “The team reached a ____ understanding, but nothing was signed”?

A. legal B. formal C. public D. mutual

56. (C1) Which tone is most accurate in “I appreciate your input, but your analysis ignores the underlying economic factors”?

A. playful B. dismissive but professional C. confused and panicked D. amused and casual

57. (C1) What does “correlation does not imply causation” mean in academic writing?

A. Co-occurrence proves cause.

B. Related variables may not cause one another.

C. Nothing is related.

D. Statistics are always wrong.

58. (C1) Which sentence contains understatement?

A. After 36 hours without sleep, he said, “I am a bit tired.”

B. I could eat ten horses.

C. The thunder was louder than a plane.

D. The fire spread quickly.

59. (C1) Which paraphrase best captures “Her leadership was instrumental in preventing the project from collapsing”?

A. She was the main reason the project survived.

B. She caused the project to fail.

C. She was not important.

D. She ignored the project.

60. (C1) What is likely implied by "The company has decided to 'restructure,' effective immediately"?

A. raises B. layoffs or major changes C. vacation D. nothing important

SECTION D. MORPHOLOGICAL COMPETENCE (Items 61–80)

Focus: word formation, affixes, derivation, inflection.

61. (B1) What is the base word in "unhappiness"?

A. un- B. happy C. -ness D. happiness

62. (B1) Which adjective best completes: "He is a very ____ person. He always donates time to help others"?

A. generosity B. generous C. generousness D. generate

63. (B1) Which word means "able to be believed"?

A. believable B. believe C. believing D. unbelief

64. (B1) Which past form correctly completes: "Yesterday she ____ her keys on the bus"?

A. lose B. losing C. lost D. loses

65. (B1) Which pair shows a verb-to-noun derivation?

A. decide → decision B. happy → happiness C. strong → strongly D. create → creative

66. (B2) Which word best completes: "The manager emphasized the need for greater ____ in reporting expenses"?

A. transparent B. transparency C. transparently D. transparentness

67. (B2) Which prefix forms the opposite of "legal"?

A. re- B. pre- C. il- D. sub-

68. (B2) Which adjective best completes: "The speaker delivered a highly ____ message that motivated the entire audience"?

A. inspiration B. inspire C. inspiredly D. inspiring

69. (B2) Which verb form best completes: "We must ____ the proposal before submitting it to the board"?

A. clarification B. clarifier C. clarify D. clarifiedness

70. (B2) Which noun correctly names the state/quality of "rely"?

A. reliness B. reliance C. relyment D. reliosity

71. (B2) What is the function of "-ment" in "enforcement"?

A. shows tense B. turns a verb into a noun C. shows plural D. turns a noun into a verb

72. (B2) Which negative form of "comfortable" is correct in formal writing?

A. discomfort B. uncomfort C. discomfortable D. discomfortable

73. (B2) Which form names a person who does the action: "to research → ____"?

A. researchness B. researcher C. researching D. researchedness

74. (B2) Which adjective best completes: "This policy is both reasonable and ____, which is why it gained support"?

A. implement B. implementation C. implementable D. implemented

75. (B2) Which noun is the best academic transformation of "refuse"?

A. refusion B. refusal C. refusize D. refusement

76. (C1) Which morphological breakdown of "misinterpretation" is correct?

A. mis + interpret + ation

B. mis + in + terpret + tion

C. misin + terpret + tion

D. mis + inter + pretation

77. (C1) Which sentence uses nominalization appropriately?

A. The experiment failed, which was bad.

B. The failure of the experiment raised ethical concerns.

C. The experiment failed and it was failing.

D. The experiment was a fail that failed.

78. (C1) Which adjective best completes: "The results were statistically ____"?

A. significantity B. significant C. signifying D. significance

79. (C1) Which revision shows precise morphological control of "safe/safety"?

A. make Internet use safety

B. increase the safety of Internet use

C. make Internet use safeness

D. safety the Internet

80. (C1) Which derived adjective best fits a formal tone: "Her comments were both ____ and respectful"?

A. constructive B. construction C. construct D. constructedly

SECTION E. ORTHOGRAPHIC COMPETENCE (Items 81–100)

Focus: spelling, capitalization, punctuation, mechanics.

81. (B1) Which spelling is correct?

A. enviroment B. environment C. envirement D. enviroement

82. (B1) Which sentence uses capitalization correctly?

A. I am studying psychology at Central State University.

B. I am studying Psychology at central state university.

C. I am studying psychology at central state university.

D. I am studying Psychology at Central state University.

83. (B1) Which sentence is punctuated correctly for a simple series?

A. I bought apples, oranges and, bananas.

B. I bought apples oranges, and bananas.

C. I bought apples, oranges, and bananas.

D. I bought, apples, oranges and bananas.

84. (B1) Which option shows correct apostrophe use for singular possession?

A. The students locker was broken.

B. The student's locker was broken.

C. The students' locker was broken, it belongs to only one student.

D. The students locker' was broken.

85. (B1) Which sentence punctuates direct speech correctly?

A. “I am tired.” she said.

B. “I am tired”, she said.

C. “I am tired,” she said.

D. “I am tired.”, she said.

86. (B2) Which sentence places the comma correctly after an introductory clause?

A. After reviewing the data the committee reached a decision.

B. After reviewing the data, the committee reached a decision.

C. After reviewing the data the committee, reached a decision.

D. After reviewing, the data the committee reached a decision.

87. (B2) Which spelling is correct?

A. occassion B. ocassion C. occasion D. occation

88. (B2) Which sentence capitalizes titles/institutions correctly?

A. The Dean of Students will meet with the president of the psychology club.

B. The dean of students will meet with the President of the Psychology Club.

C. The Dean of students will meet with the President of the Psychology club.

D. The dean of Students will meet with the president of the Psychology Club.

89. (B2) Which sentence uses a semicolon correctly?

A. The study was complex; however, the findings were clear.

B. The study was complex; because the findings were clear.

C. The study was complex; and the findings were clear.

D. The study was complex; the findings were, clear.

90. (B2) Which sentence hyphenates a compound adjective correctly before a noun?

A. a well researched-report

B. a well-researched report

C. a well researched report

D. a well-research report

91. (B2) Which sentence punctuates a series of three items correctly?

A. planning, funding and, community support

B. planning, funding, and community support

C. requires, planning, funding, and community support

D. planning funding, and community support

92. (B2) Which option best corrects the comma splice: “The survey was anonymous, the responses were honest”?

A. The survey was anonymous; the responses were honest.

B. The survey was anonymous, but, the responses were honest.

C. The survey was anonymous the responses were honest.

D. The survey was anonymous: and the responses were honest.

93. (B2) Which sentence uses quotation punctuation correctly (AmE)?

A. …is “a double-edged sword”.

B. …is “a double-edged sword.”

C. …is, “a double-edged sword”.

D. …, that social media is “a double-edged sword”.

94. (B2) Which sentence shows consistent tense and correct mechanics?

A. She emails the professor yesterday…

B. She emailed the professor yesterday…

C. She email the professor yesterday…

D. She emailing the professor yesterday…

95. (B2) Which sentence uses the apostrophe correctly for plural possession?

A. The students' concerns were addressed by the dean.

B. The student's concerns were addressed … referring to a group.

C. The students concerns' were addressed…

D. The students concern's were addressed…

96. (C1) Which sentence uses an em dash correctly for emphasis?

A. The results were clear — the treatment group improved significantly.

B. The results were — clear the treatment group improved significantly.

C. The results were clear the treatment group — improved significantly.

D. The results — were clear the treatment group improved significantly.

97. (C1) Which sentence uses a colon correctly to introduce a list?

A. The participants reported three main sources of stress: exams, finances, and sleep deprivation.

B. The participants reported: three main sources of stress…

C. The participants: reported three main sources of stress…

D. …three main sources of stress, exams: finances, and sleep deprivation.

98. (C1) Which sentence punctuates two independent clauses most appropriately for academic writing?

A. Anxiety affects sleep it also affects concentration.

B. Anxiety affects sleep; it also affects concentration.

C. Anxiety affects sleep, it also affects concentration.

D. Anxiety affects sleep also affects concentration.

99. (C1) Which sentence correctly sets off nonessential information with commas?

A. The professor, who has published five books, will lead the seminar.

B. The professor who has published five books will lead the seminar.

C. The professor who, has published five books will lead the seminar.

D. The professor who has published, five books will lead the seminar.

100. (C1) Which sentence shows final-edit quality in mechanics, clarity, and tone?

A. proves … bad for everyone…

B. kind of shows … pretty harmful

C. suggests that excessive social media use may negatively affect concentration

D. totally destroying attention spans

SECTION F. PHONOLOGICAL COMPETENCE (Items 101–120)

Focus: sound discrimination, word and sentence stress, rhythm, linking, intonation, and phoneme–grapheme correspondence.

(B1 Level — Items 101-105) Basic sound and stress recognition

101. (B1) Which pair of words contains the same vowel sound?

A. ship – sheep B. sit – seat C. pen – ten D. full – fool

102. (B1) Which word has a different initial sound?

A. cat B. cup C. city D. cold103.

103. (B1) Which syllable is stressed in *computer*?

A. COM B. PU C. TER D. None

104. (B1) Which word has the same final sound as *leaf*?

A. safe B. give C. laugh D. leave

105. (B1) Which pronunciation pattern is correct for the plural *books*?

A. /-s/ B. /-z/ C. /-ɪz/ D. none

(B2 Level — Items 106-113) Connected speech and stress rules

106. (B2) Which word has stress on the first syllable?

A. record (noun) B. record (verb) C. relax D. begin

107. (B2) Which word has the same vowel sound as *caught*?

A. cot B. court C. cat D. cut

108. (B2) Which sentence shows correct intonation for a yes/no question?

A. Falling tone at the end B. Rising tone at the end C. Flat tone D. Rise–fall tone

109. (B2) In natural connected speech, how is the phrase **"next year"** commonly pronounced?

A. [neks jɪər] with the /t/ sound softened or blended

B. [nekt jɪər] with a strongly pronounced /t/

C. [nexə jɪər] with an added schwa

D. [nɛk st jɪər] with a clear break between the words

110. (B2) Which syllable is stressed in *photography*?

A. PHO B. TO C. GRA D. PHY

111. (B2) Which minimal-pair contrast depends on voicing?

A. fan – van B. pin – pen C. ship – sheep D. coat – caught

112. (B2) In the word *walked*, the final –ed is pronounced _____.

A. /id/ B. /t/ C. /d/ D. /ed/

113. (B2) Which word contains a silent letter?

A. write B. rain C. rose D. rightly

(C1 Level — Items 114-120) Suprasegmental features and phonological analysis

114. (C1) Which sentence would likely have a falling intonation?

A. Where are you going? B. How beautiful this is! C. Would you like some coffee? D. Are you coming later?

115. (C1) In fluent speech, *"black coffee"* is often pronounced as:

A. [blæk kɒfi] B. [blæ kɒfi] C. [blæ kɔfi] D. [blæk ɔfi]

116. (C1) Which word pair shows difference in primary stress pattern?

A. 'import (noun)' vs im'port (verb) B. teacher vs teaching C. student vs study D. open vs opened

117. (C1) Which term refers to the reduction of unstressed vowels to /ə/?

A. assimilation B. elision C. linking D. schwa-reduction

118. (C1) In the sentence *"He didn't call, did he?"* the tag question has a ____ intonation when the speaker only wants confirmation.

A. rising B. falling C. rise-fall D. flat

119. (C1) Which process describes how sounds influence each other at boundaries (e.g., *good boy* → /gʊb bɔɪ/)?

A. elision B. assimilation C. intrusion D. reduction

120. (C1) Which pronunciation distinction differentiates *ship* /ɪ/ and *sheep* /iː/?

A. vowel length and quality B. consonant voicing C. stress placement D. syllable number

MASTER ANSWER KEY

Lexical (1–20): 1 B, 2 B, 3 B, 4 C, 5 C, 6 B, 7 B, 8 D, 9 B, 10 C, 11 C, 12 A, 13 A, 14 C, 15 A, 16 B, 17 D, 18 C, 19 A, 20 A

Syntactic (21–40): 21 B, 22 C, 23 B, 24 B, 25 C, 26 B, 27 C, 28 A, 29 B, 30 C, 31 C, 32 B, 33 C, 34 B, 35 C, 36 B, 37 A, 38 C, 39 B, 40 B

Semantic (41–60): 41 C, 42 C, 43 B, 44 A, 45 C, 46 B, 47 C, 48 C, 49 A, 50 B, 51 C, 52 B, 53 B, 54 B, 55 D, 56 B, 57 B, 58 A, 59 A, 60 B

Morphological (61–80): 61 B, 62 B, 63 A, 64 C, 65 A, 66 B, 67 C, 68 D, 69 C, 70 B, 71 B, 72 A, 73 B, 74 C, 75 B, 76 A, 77 B, 78 B, 79 B, 80 A

Orthographic (81–100): 81 B, 82 A, 83 C, 84 B, 85 C, 86 B, 87 C, 88 A, 89 A, 90 B, 91 B, 92 A, 93 B, 94 B, 95 A, 96 A, 97 A, 98 B, 99 A, 100 C

Phonological Competence (101-120): 101 C, 102 C, 103 B, 104 A, 105 A, 106 A, 107 B, 108 B, 109 A, 110 C, 111 A, 112 B, 113 A, 114 B, 115 A, 116 A, 117 D, 118 B, 119 B, 120 A

C- Map of the Research Locale (Include Coordinates)

D- Narrative & Photo Documentation
E- Ethics Committee Approval

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